

Normal Approximation to the Binomial Distribution

Example

A factory produces electronic components, and historically 60% of the components are defect-free. Suppose 200 components are selected randomly. Find the probability that at least 130 components are defect-free, using the normal approximation to the binomial distribution.

Solution

Let X be the number of defect-free components.

$$X \sim \text{Bin}(n=200, p=0.6)$$

Check whether the normal approximation is appropriate:

$$np = 200(0.6) = 120 > 5$$

$$n(1-p) = 200(0.4) = 80 > 5$$



Since both conditions are satisfied
 $\Rightarrow$ Normal approximation

Since both np and $n(1 - p)$ are greater than 5, the normal approximation is appropriate.

Step 1: Find the mean and standard deviation

For a binomial distribution:

$$\mu = np$$

$$\mu = 200(0.6) = 120$$

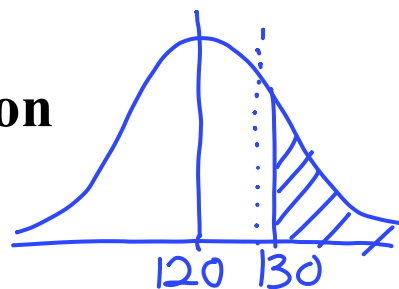
$$\sigma = \sqrt{np(1 - p)}$$

$$\sigma = \sqrt{200(0.6)(0.4)} = \sqrt{48}$$

$$Y \sim N(\mu=120, \sigma=6.928)$$

Step 2: Apply the continuity correction

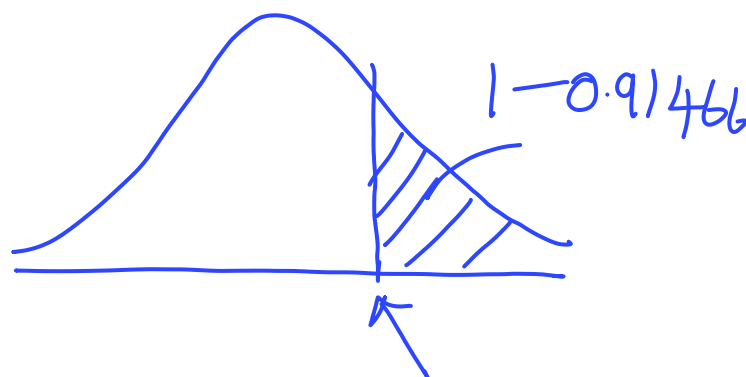
We want to find: $P(X \geq 130)$



Using the continuity correction, the discrete value 130 is represented by the interval from 129.5 to 130.5.

Therefore:

$$P(X \geq 130) \approx P(Y > 129.5)$$



Step 3: Standardize

$$Z = (Y - \mu) / \sigma$$

$$P(Y > 129.5) = P[Z > (129.5 - 120) / 6.928] = 1.37$$

Step 4: Use the standard normal table

From the standard normal table:

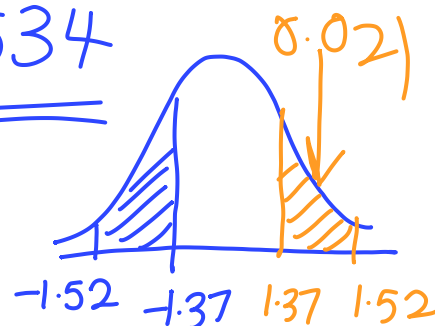
$$P(Z < 1.371) \approx 0.08534$$

Therefore:

$$P(Z > 1.371) =$$

$$P(X \geq 130) \approx 0.08534$$

$$P(X=110) \approx \underline{\underline{0.021}}$$



$$P(X=110) = ? \quad P(109.5 \leq X \leq 110.5)$$

Final Answer

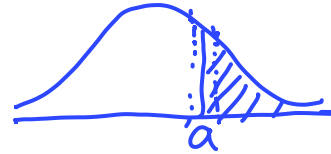
after cont. corr.

$$P(X \geq 130) \approx 0.0852$$

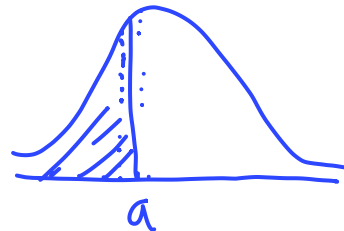
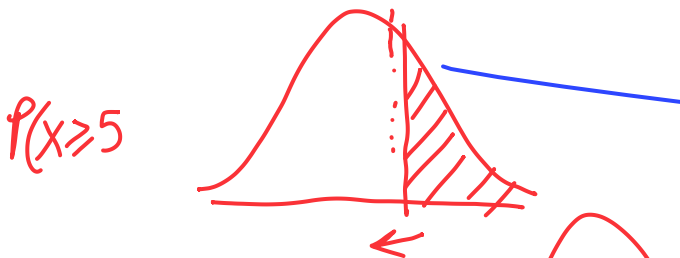
$$P(-1.52 \leq Z \leq -1.37)$$

Thus, there is approximately an 8.5% probability that at least 130 of the 200 components will be defect-free.

Continuity Correction Guide



Binomial Probability	Normal Approximation
$P(X \geq a)$	$P(Y > a - 0.5)$
$P(X > a)$	$P(Y > a + 0.5)$
$P(X \leq a)$	$P(Y < a + 0.5)$
$P(X < a)$	$P(Y < a - 0.5)$
* $P(X = a)$	$P(a - 0.5 < Y < a + 0.5)$
$P(a \leq X \leq b)$	$P(a - 0.5 < Y < b + 0.5)$



True Bin. pro

$${}^n C_x \theta^x (1-\theta)^{n-x} = \frac{200!}{110! 90!} (0.6)^{110} (0.4)^{90} =$$